

Ivan Augusto Xavier Souza

Senior Backend Engineer

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Professional Summary

Senior Software Engineer with 17+ years of experience architecting and building distributed systems at scale for government and enterprise clients. Expert in **backend architecture (Node.js, Go)**, **microservices**, **cloud-native platforms (Kubernetes/OpenShift)**, and **CI/CD pipelines with integrated security gates**. Proven track record in modernizing legacy systems, migrating monoliths to microservices, and reducing delivery cycles through automation. Red Hat Champion. Author of open-source tools for BLE, SOLID ID generation, and security hardening.

Technical Skills

Languages: TypeScript, JavaScript, Go, PHP, Rust

Backend & APIs: Node.js, Express, NestJS, Go, RESTful APIs, gRPC, WebSockets, SSE, event-driven architecture

Cloud & Platform: Kubernetes, OpenShift, Docker, Docker Compose, Terraform, Ansible

CI/CD & Security: Jenkins, GitLab CI/CD, GitHub Actions, Azure DevOps, SAST/DAST, automated code review

Databases & Storage: PostgreSQL, MongoDB, Redis, Oracle, MySQL/MariaDB, OpenSearch, RabbitMQ

Professional Experience

Senior Backend Engineer | CENTRAL IT TECNOLOGIA DA INFORMAÇÃO S/A

Jul 2023 – Jan 2026 | Brasília, Brazil

Architected and led the re-engineering of an enterprise chatbot platform from a monolithic NestJS codebase to a distributed microservices architecture in Go, serving government clients including SERPRO.

- **Architected the migration** of a NestJS monolith into **5+ independently deployable microservices in Go**, running on Kubernetes with autoscaling and high availability
- **Built a complete CI/CD pipeline** on Jenkins with SAST, DAST, and automated code review gates — replacing manual bi-weekly deployments with on-demand, gated releases

- **Developed Rust components** for the platform's NLU engine, processing high-throughput intent classification for chatbot conversations
- **Created an embeddable web widget** for chatbot integration into legacy government portals, deployed across multiple client websites
- **Established architecture standards** — service contracts, testing strategies, code review guidelines, and observability patterns — for a cross-functional team of 10+ engineers

Technologies: Go, Node.js, NestJS, Rust, TypeScript, PostgreSQL, Kubernetes, Jenkins, SAST/DAST

Tech Lead & Scrum Master | CENTRAL INFORMATION TECHNOLOGY LTDA.

Sep 2022 – Mar 2023 | Brasília, Brazil

Led a 3-person engineering team building a BLE beacon monitoring mobile application with IoT capabilities.

- **Defined the architecture** for a React Native mobile app with AdonisJS/Node.js backend and MongoDB, balancing time-to-market with long-term maintainability
- **Built an IoT module** integrating Serial and BLE communication, expanding the product into physical asset tracking
- **Established code review culture** and mentored junior developers, raising team code quality and consistency

Technologies: React Native, Node.js, AdonisJS, MongoDB, BLE, Serial

Solutions Architect | ALLTECH SOLUÇÕES EM TECNOLOGIA LTDA.

Oct 2019 – May 2022 | Brasília, Brazil

Red Hat Champion leading architecture and deployment of enterprise Kubernetes/OpenShift solutions for clients including Infraero (national airport operator, 54+ airports).

- **Designed and deployed OpenShift clusters** with up to **30 nodes**, implementing high-availability control planes, persistent storage via Ceph, and Day-2 operations automated with Ansible
- **Automated infrastructure provisioning** with Ansible, reducing setup time and standardizing configurations across environments
- **Conducted technical pre-sales and OpenShift demonstrations** for enterprise clients, translating business requirements into architecture proposals
- **Migrated applications to containers** with CI/CD pipelines in Jenkins and GitLab CI

Technologies: Kubernetes, OpenShift, Ansible, Docker, Ceph, Jenkins, GitLab CI, ESXi, vCenter

Systems Analyst | INDRA SISTEMAS S/A (CAPES)

Jul 2018 – Jun 2019 | Brasília, Brazil

- **Built REST APIs with JWT authentication** for large-scale government education platforms using Laravel/Lumen and Angular, serving the Brazilian federal education agency (CAPES)
- **Integrated Redis and Memcached** for high-traffic endpoint caching, reducing database load during peak usage
- **Set up CI/CD with Docker, Jenkins, and OpenShift**, enabling automated builds and containerized deployments

Technologies: PHP, Laravel, Lumen, Angular, Oracle, Redis, Memcached, Docker, Jenkins, OpenShift

Systems Analyst | INDRA SISTEMAS S/A (Ministry of Education)

Jan 2017 – Jul 2018 | Brasília, Brazil

- **Architected the SisProuni 2.0 modernization** — Brazil's national university scholarship program — defining REST API contracts with HAL convention and transforming a legacy monolithic system
- **Contributed to the Carolina Bori Platform** — national degree validation system — using Scrum methodology

Technologies: PHP, Zend Framework, jQuery, Oracle

Founder & Software Engineer | NAESQUINA SOLUÇÕES EM TECNOLOGIA LTDA.

Mar 2011 – Present | Brasília, Brazil

Building proprietary solutions in geolocation search, real-time vehicle tracking, and AI-powered automation.

- **Built a real-time tracking and messaging platform** with Node.js, RabbitMQ, WebSockets, and OpenSearch, processing geolocation data for mobile fleets
- **Designed 100% containerized infrastructure** on Kubernetes with Docker, PostgreSQL, and Redis
- **In development: AI sales agent** for automating customer engagement and lead qualification

Technologies: Node.js, Angular, Ionic, PostgreSQL, RabbitMQ, Redis, WebSockets, OpenSearch, Kubernetes, Docker

Additional Experience

Period	Role	Company
Aug 2015 – Jan 2017	Software Engineer	MBA TECNOLOGIA — 6 mobile apps for MEC, INEP, Ministry of Health
Nov 2013 – Dec 2014	Systems Analyst/Developer	SQUADRA (ANTT) — GIS railway system (PHP, PostGIS, ETL)
Oct 2008 – Oct 2010	Systems Analyst	UNIMIX — TCU/Imprensa Nacional web portals (PHP, Oracle)
Dec 2007 – Oct 2008	Junior/Mid-Level Developer	TECNISYS — Biometric auth (PalmSecure), Java, Oracle

Open Source Projects

Bluetooth Virtual Keyboard (GitHub) — Turns a Linux computer into a Bluetooth keyboard for other devices via BLE, reading kernel input events (`/dev/input/eventX`) and communicating with BlueZ over D-Bus. Built with Clean Architecture.

SOLID ID (GitHub) — 128-bit unique ID generator (timestamp + entropy + CRC-16) encoded in 22 Base62 characters. Time-sortable, checksum-validatable, zero dependencies.

Debian 13 Hardening & Tuning (GitHub) — Ansible playbook for security hardening (CIS/STIG) and performance tuning of Debian 13 servers. 50+ customizable variables.

k8s-one (GitHub) — Single-node Kubernetes cluster built from scratch with official upstream components — no K3s, no KIND, no kubeadm. Includes etcd, containerd, Calico, CoreDNS, and HAProxy in a distroless image.

Education & Certifications

Computer Science (B.S.) — Cruzeiro do Sul University (*in progress — expected graduation: 2029*)

Certified ScrumMaster (CSM) — Scrum Alliance, 2016

Languages

Portuguese: Native | **English:** Intermediate — strong technical reading comprehension, professional written communication